

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

FRIDAY, 20 MAY 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL THREE (3)** questions in **Section A** and choose **ONLY ONE (1)** question in **Section B**.

Should a candidate answer more than **ONE (1)** question in section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Section A [Total: 75 marks]**

This section consists of **THREE (3) COMPULSORY** questions.

- Q1. (a) The following is part of the Budi Clinic relational scheme which shows the relationship between doctor, patient, treatment, medication and treatment prescription. Suppose a patient does not see the same doctor more than once in a day. For each treatment, several medications will be prescribed to the patient. Dates are recorded as dd/mm/yyyy (day, month and year). The primary keys are highlighted in bold.

Patient (**patientNo**, pFirstName, pLastName, pAddress, pDOB, pPhoneNo)

Doctor (**doctorID**, docFirstName, docLastName, docStartDate, docExpertise, docSalary)

Medicine (**medCode**, medName, medDescription, medQuantity, medExpiryDate, medUnitPrice)

Treatment (**treatID**, patientNo, doctorID, treatDate, treatType, treatPayment)

Prescription (**treatID**, **medCode**, presQuantity, presTotalPrice)

- (i) Write **ONE (1)** Structural Query Language (SQL) statement to list the names, types of treatment and names of medications prescribed for all patients who received treatment between 1 December 2021 and 31 January 2022. (8 marks)
- (ii) For each area of expertise that has more than two doctors, write **ONE (1)** SQL statement to list the number of doctors in that area of expertise. Label it as "Number of Expertise". (5 marks)
- (iii) Write **ONE (1)** SQL statement to list all the names of doctors who have expertise in the field of 'Radiology' as well as the number of treatments given throughout the year 2021. Sort the dates of treatment from earliest to most recent. (8 marks)
- (iv) Write **ONE (1)** SQL statement to show the new salary of doctor with id "D15045" which has been increased by 10% and rename the column as "Salary Increment". (4 marks)

[Total : 25 marks]

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Q2. (a) Figure 2.1 shows the relations of Account Transaction database.

Table name: Customer			Table name: Account	
CustomerID	Name	TelNo	AccountNo	Balance
C1356	Choo	012-5399183	AC0567	\$12000
C8843	Sally	011-3918475	AC4918	\$750
C3467	Adrian	010-5729192	AC9132	\$9000
			AC1223	\$200
			AC1784	\$15500

Table name: Transaction			
CID	AccNo	TransDate	TransAmount
C1356	AC0567	21/01/2022	\$8000
C1356	AC4918	01/02/2022	\$3000
C8843	AC9132	15/03/2022	\$100
C3467	AC1223	17/02/2022	\$450
C3467	AC1784	23/01/2022	\$5000

Figure 2.1 Partial Records in Account Transaction Database

- (i) Identify all parent and child tables according to Figure 2.1. (3 marks)
- (ii) Based on the answers in Q2(a)(i), identify the strength of the relationship that exists. (2 marks)
- (iii) Suggest the appropriate data type for *customerID* and *Balance* attributes from Customer and Account tables respectively. (2 marks)
- (iv) Identify the appropriate action to be taken in the Transaction table if account number *AC1223* is closed. Justify your answer. (3 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q2. (Continued)**

- (b) The ERD in Figure 2.2 shows the project involved by the employees in the company.

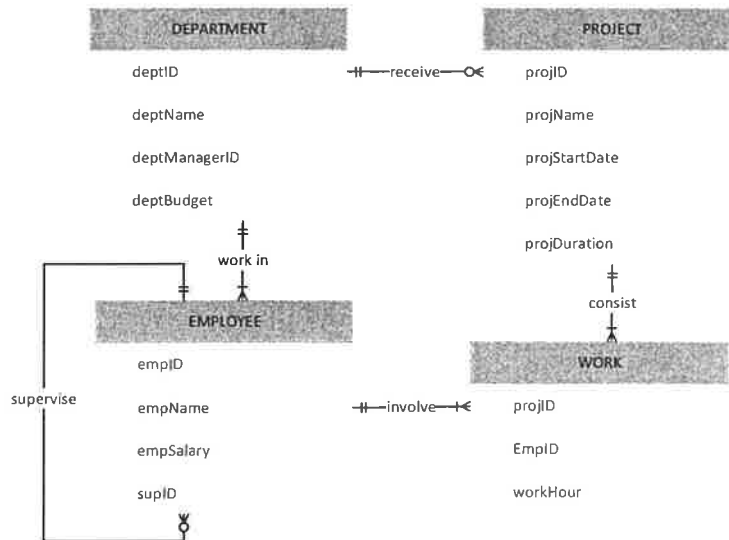


Figure 2.2 The partial entity relationship diagram (ERD) of Employee Database

- (i) Identify the primary key and the foreign key(s) for Employee and Project entities. If the entity does not have a foreign key, write NONE. (5 marks)
- (ii) Identify **ONE (1)** derived attribute and specify how the value is derived. (3 marks)
- (iii) Identify **ONE (1)** weak entity. Justify your answer. (3 marks)
- (iv) Explain how the entity integrity and referential integrity rules are applied to these relations. (4 marks)

[Total : 25 marks]

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- Q3. Figure 3.1 shows the sample information of club activities at a secondary school. Suppose each student is allowed to join only one club.

StudentName	StudentID	Classroom	Club	ClubLeader	ClubDay	Advisor
Emma	S001	1B	C01 Archery	Derrick	Wednesday	Mr. Adam (T045) Mr. Philip (T035)
Mia	S002	3A	C02 Sewing	Sally	Monday	Ms. Hanna (T014) Ms. Nicole (T003)
Lucas	S003	2A	C03 Tennis	Benjamin	Friday	Mr. Smith (T023)
Sophia	S004	3C	C01 Archery	Derrick	Wednesday	Mr. Adam (T045) Mr. Philip (T035)
James	S005	4E	C04 Badminton	Noah	Friday	Mr. Richard (T19)
Derrick	S006	5B	C01 Archery	Derrick	Wednesday	Mr. Adam (T045) Mr. Philip (T035)

Figure 3.1 Sample Records for Club Activities

- Based on Figure 3.1, identify whether the relation is in a First Normal Form (1NF) or not. Justify your answer. (3 marks)
 - Based on Figure 3.1, identify **THREE (3)** anomalies in the records. Provide an example for each anomaly. (6 marks)
 - Identify **TWO (2)** rules to convert First Normal Form (1NF) to Second Normal Form (2NF). (2 marks)
 - Draw the dependency diagram showing all dependencies in the relation, based on the sample data shown in Figure 3.1. (5 marks)
 - Normalize the relation to Third Normal Form (3NF). Label the primary key and foreign key in each relation. (9 marks)
- [Total : 25 marks]

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Section B [Total: 25 marks]**

This section consists of **TWO (2) OPTIONAL** questions. Answer **ONE (1)** question only.

- Q4. (a) Figure 4.1 shows a student entity in which a student can tutor one or more other students. Contrary, that student may also be tutored by one or more other students.

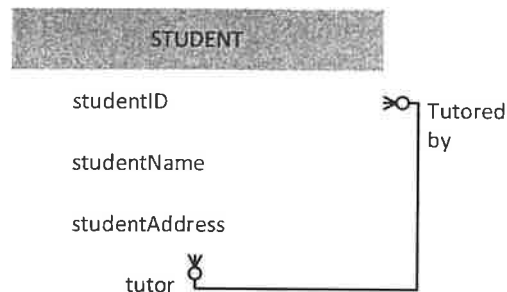


Figure 4.1. Student Entity

- (i) Map the data model shown in Figure 4.1 to a set of relational schemas. (2 marks)
 - (ii) Based on the answers in Q4(a)(i), identify the primary key and foreign key(s) in each table. (4 marks)
- (b) Experiments in a laboratory need to be done by students using one or more specific materials. A student must make one or more experiments. An experiment can be done individually or in group with no more than five members. Student information such as student number, name, class and guardian phone number should be recorded. An experiment must be completed within a certain period of time. Each experiment was given a unique identification number and could be performed by more than one group of students. The name of each material is unique and can be used in more than one experiment.
- (i) Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least **THREE (3)** relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD. (15 marks)
 - (ii) Based on the answers in Q4(b)(i), identify **ONE (1)** strong entity and **ONE (1)** strong relationship. Justify your answer. (4 marks)
- [Total : 25 marks]

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- Q5. (a) Explain the differences between the following pairs of attributes and give an example for each of it.
- (i) Simple attribute and composite attribute. (4 marks)
 - (ii) Single-valued attribute and multi-valued attribute. (4 marks)
- (b) Although conceptual model can handle many-to-many relationship, however this type of relationship should not be implemented in a relational DBMS. Suggest **ONE (1)** solution to overcome this problem. (2 marks)
- (c) AtoZ Enterprise is an employment agency that provides employment for prospective employers. For that purpose, the agency needs to keep details such as name, identity card number, address, telephone number, date of birth, and skill type of the candidate applying for the job. Each candidate can include up to four types of skills when applying for a job. In addition, the agency also needs to keep the employer-related data. Requests for employees are submitted by the employer to the agency by stating the type of work, the salary offered and the date the employees is required. The agency will then look for a number of qualified candidates to go through the interview process.

Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least **THREE (3)** relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD.

(15 marks)

[Total : 25 marks]

